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CASTING APPARATUS AND CASTING METHOD

Abstract

The casting apparatus includes a movable mold and a fixed mold, and includes a mold having a coolant channel therein, a mold clamping device, a coolant pump, a laser scanner for detecting an outer shape of the die-cast product, and a control unit, wherein the processor of the control unit acquires an outer shape of the die-cast product from the laser scanner when the mold is separated after casting, calculates a deformation amount with respect to a design shape of the die-cast product based on the acquired outer shape, and adjusts a mold clamping holding time or a coolant flow rate based on the calculated deformation amount.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-031546 filed on Mar. 1, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The disclosure relates to a structure of a casting apparatus and a casting method using the casting apparatus.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2003-191065 (JP 2003-191065 A) discloses a casting apparatus capable of efficiently replacing a dedicated part by drawing out the dedicated part from a general-purpose part and attaching a new dedicated part.

SUMMARY

[0004] Now, when die-cast products are consecutively cast using the casting apparatus such as described in JP 2003-191065 A, deformation of the die-cast product may become great. However, in the casting apparatus according to the related art, feedback of deformation amount to casting conditions cannot be performed, since the deformation amount with respect to a designed shape of the die-cast product is not detected. Accordingly, in the casting apparatus according to the related art, suppressing increase in the deformation amount when performing consecutive casting is difficult.

[0005] Accordingly, an object of the present disclosure is to suppress increase in the deformation amount of a die-cast product when performing consecutive casting of the die-cast product.

[0006] The casting apparatus according to the present disclosure is [0007] a casting apparatus that performs casting of a die-cast product, the casting apparatus including [0008] a mold made up of a movable mold and a fixed mold, within which is provided a coolant channel, [0009] a mold clamping device for performing mold clamping and separation of the mold, [0010] a coolant pump for causing coolant to flow over the coolant channel of the mold, [0011] a shape detector for detecting an external shape of the die-cast product, and [0012] a control unit for adjusting mold clamping holding time of the mold by the mold clamping device or a coolant flow rate of the coolant pump, in which [0013] the control unit includes a processor for performing information processing, and the processor [0014] acquires the external shape of the die-cast product from the shape detector when separating the mold following casting, [0015] calculates a deformation amount with respect to a designed shape of the die-cast product, based on the external shape of the die-cast product that is acquired, and [0016] adjusts the mold clamping holding time or the coolant flow rate based on the deformation amount that is calculated.

[0017] In this way, the mold clamping holding time or the coolant flow rate is adjusted in accordance with the deformation amount with respect to the design shape of the die-cast product following casting, and accordingly increase in the deformation amount of the die-cast product can be suppressed when performing consecutive casting.

[0018] In the casting apparatus according to the present disclosure, the processor may increase the coolant flow rate when the deformation amount that is calculated exceeds a predetermined threshold value.

[0019] Accordingly, rise in the temperature of the die-cast product when performing consecutive casting can be suppressed, and increase in the deformation amount of the die-cast product can be suppressed.

[0020] In the casting apparatus according to the present disclosure, the processor may increase the mold clamping holding time when the deformation amount that is calculated exceeds a predetermined other threshold value.

[0021] During the mold clamping holding time, the coolant is distributed to the mold, and accordingly rise in the temperature of the die-cast product when performing consecutive casting due to increasing the mold clamping holding time can be suppressed, and increase in the deformation amount of the die-cast product can be suppressed.

[0022] In the casting apparatus according to the present disclosure, the shape detector may be a laser scanner or an ultrasonic shape detector.

[0023] Thus, the deformation amount of the die-cast product can be detected non-contactlessly.

[0024] A method of casting a die-cast product according to the present disclosure includes [0025] preparing a casting apparatus that includes a mold made up of a movable mold and a fixed mold, within which is provided a coolant channel, a mold clamping device for performing mold clamping and separation of the mold, a coolant pump for causing coolant to flow over the coolant channel of the mold, and a shape detector for detecting an external shape of the die-cast product, [0026] detecting the external shape of the die-cast product by the shape detector when separating the mold following casting, [0027] calculating a deformation amount with respect to a designed shape of the die-cast product, based on the external shape of the die-cast product that is detected, and [0028] increasing a coolant flow rate of the coolant pump when the deformation amount that is calculated exceeds a predetermined threshold value, or increasing the mold clamping holding time of the mold when the deformation amount that is calculated exceeds a predetermined other threshold value.

[0029] Thus, increase in the deformation amount of the die-cast product when performing consecutive casting can be suppressed.

[0030] The present disclosure can suppress increase in the deformation amount of the die-cast product when the die-cast product is consecutively cast.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0032] FIG. 1 is a system diagram illustrating a configuration of a casting apparatus according to an embodiment;

[0033] FIG. 2 is a flow chart illustrating the operation of the casting apparatus shown in FIG. 1 in casting die-cast products;

[0034] FIG. 3 is an explanatory view showing a casting apparatus in a state in which a mold is separated after casting;

[0035] FIG. 4 is a diagram illustrating a change in the deformation amount of the die-cast product and a change in the coolant flow rate with respect to the number of times of continuous casting;

[0036] FIG. 5 is a flow chart showing another operation of the casting apparatus shown in FIG. 1 in casting die-cast products; and

[0037] FIG. 6 is a diagram illustrating a change in the deformation amount of the die-cast product and a change in the mold clamping holding time with respect to the number of times of continuous casting.

DETAILED DESCRIPTION OF EMBODIMENTS

[0038] Hereinafter, the casting apparatus **100** of the embodiment will be described with reference to the drawings. As illustrated in FIG. 1, the casting apparatus **100** includes a mold **10**, a mold clamping device **40**, an extrusion device **50**, an injection device **61**, an exhaust pipe **62**, a coolant

pump **36**, laser scanners **71** and **72**, and a control unit **80**. In the drawings, reference numerals FR, LH, UP denote the front, left, and upper sides of the casting apparatus **100**, respectively. The opposing orientations of FR, LH, UP are rearward, rightward, and downward, respectively. [0039] The mold clamping device **40** includes a fixed plate **41**, a movable platen **42**, a tie bar **43**, and a driving unit **45**. The fixed plate **41** is fixed to a distal end portion of a tie bar **43** which is a rod member. The movable platen **42** is disposed so as to face the fixed plate **41**, and moves forward along the tie bar **43** by the driving unit **45**.

[0040] A mold **10** is attached to the mold clamping device **40**. The mold **10** includes a fixed mold **20** and a movable mold **30**. A fixed mold **20** is attached to the fixed plate **41**, and a movable mold **30** is attached to the movable platen **42**. The mold clamping device **40** opens and closes the mold **10** by moving the movable mold **30** in the front-rear direction together with the movable platen **42**. Then, the movable mold **30** is moved forward and pressed against the fixed mold **20**, whereby the mold clamping of the mold **10** is performed. FIG. 1 shows a mold **10** in a clamped state by a mold clamping device **40**. When mold clamping of the mold **10** is performed, a cavity **12** is formed between the fixed mold **20** and the movable mold **30**. The cavity **12** is a portion of the internal space of the mold **10** having a shape corresponding to the shape of the die-cast product **90** (see FIG. 3). Here, the die-cast product **90** may be, for example, a vehicle body structural member.

[0041] The injection device **61** pumps the molten metal, which is the material of the die-cast product **90**, into the cavity **12** of the mold **10** as indicated by the white arrow in FIG. 1. Further, the air in the cavity **12** is exhausted from the exhaust pipe **62** by a vacuum device (not shown) (see an arrow in FIG. 1). The extrusion device **50** includes an extrusion pin **51**, an extrusion plate **52**, and an extrusion driving unit **55**. The extrusion device **50** separates the die-cast product **90** from the movable mold **30** by projecting the extrusion pins **51** when the mold **10** is separated.

[0042] The fixed mold **20** includes a main mold **21** and an insert **25**. The movable mold **30** includes a main mold **31** and an insert **35**. Here, the inserts **25** and **35** are portions of the fixed mold **20** and the movable mold **30** that are replaced according to the die-cast product **90** to be molded. The main molds **21** and **31** are portions fixed to the fixed plate **41** and the movable platen **42** of the mold clamping device **40**, and are portions commonly used for the die-cast product **90** to be molded.

[0043] A coolant channel **32** is provided inside the insert **35** of the movable mold **30**. In the coolant channel **32**, the coolant pump **36** and the coolant cooler **37** are connected by a coolant pipe **38**. The coolant pressurized by the coolant pump **36** flows into the coolant channel **32** of the insert **35**. The coolant whose temperature has risen in the coolant channel **32** is cooled by the coolant cooler **37** and circulated to the coolant pump **36**. In this way, the coolant pump **36** allows the coolant to flow through the coolant channel **32** of the insert **35**.

[0044] Laser scanners **71** and **72** are disposed slightly behind the fixing plate **41**. The laser scanners **71** and **72** are shape detectors that irradiate the die-cast product **90** (see FIG. 3) with laser light and detect the outer shape of the die-cast product **90** by reflected waves. The laser scanner **71** detects an outer shape of an upper portion of the die-cast product **90**, and the laser scanner **72** detects an outer shape of a lower portion of the die-cast product **90**.

[0045] The control unit **80** is a computer including a CPU **81** that is a processor that performs information processing, and a memory **82** that stores a control program and control data. The driving unit **45** of the mold clamping device **40**, the extrusion driving unit **55** of the extrusion device **50**, the injection device **61**, and the coolant pump **36** are connected to the control unit **80** and operate according to an instruction from the control unit **80**. Further, the laser scanners **71** and **72** are connected to the control unit **80**. The detected external shape data of the die-cast product **90** is input to the control unit **80**. As described with reference to FIGS. 2 to 6, the control unit **80** adjusts the mold clamping holding time T by the mold clamping device **40** or the coolant flow rate Q of the coolant pump **36**.

[0046] Next, with reference to FIGS. 2 to 4, the operation of the casting apparatus **100** when casting the die-cast product **90** will be described. As shown in FIG. 1, a mold **10** is set in the

casting apparatus **100**.

[0047] As shown in step **101** of FIG. 2, CPU **81** of the control unit **80** moves the movable mold **30** forward together with the movable platen **42** by operating the driving unit **45** of the mold clamping device **40**, the mold clamping of the mold **10** is pressed against the fixed mold **20**. Then, in step **102** of FIG. 2, CPU **81** of the control unit **80** operates the injection device **61** to inject the molten metal into the cavity **12** of the mold **10**. Then, in step **103** of FIG. 2, CPU **81** operates the coolant pump **36** to cause the coolant to flow through the coolant channel **32**. Then, CPU **81** maintains the duty of the coolant pump **36** at a predetermined value, maintains the coolant flow rate Q at a predetermined flow rate, and maintains the mold clamping state. CPU **81** waits until the predetermined mold clamping holding time T elapses, as shown in step **104** of FIG. 2. Then, when the predetermined mold clamping holding time T has elapsed, CPU **81** determines YES in step **104** of FIG. 2 and proceeds to step **105** of FIG. 2.

[0048] In step **105** of FIG. 2, CPU **81** operates the driving unit **45** of the mold clamping device **40** as shown in FIG. 3 to move the movable mold **30** backward together with the movable platen **42**, and separates the mold **10** into the fixed mold **20** and the movable mold **30**. When the mold **10** is separated after casting, a die-cast product **90** is formed in the region of the cavity **12** shown in FIG. 1 (see FIG. 3). Further, when the mold **10** is separated into the fixed mold **20** and the movable mold **30**, the laser beam from the laser scanners **71** and **72** can reach the surface of the die-cast product **90**. In step **105** of FIG. 2, CPU **81** operates the extrusion driving unit **55** of the push-out device **50** to move the push-out pin **51** forward to release the die-cast product **90** from the movable mold **30**.

[0049] In step **106** of FIG. 2, CPU **81** detects the outer shape of the die-cast product **90** by the laser scanners **71** and **72**, and acquires data of the detected outer shape from the laser scanners **71** and **72**. At this time, CPU **81** acquires the external shapes of the outer peripheral portions **91** and **92** of the die-cast product **90** in which the deformation Δ from the designed shape increases. Then, in step **107** of FIG. 2, CPU **81** compares the outer shape of the die-cast product **90** acquired from the laser scanners **71** and **72** with the design shape, and calculates the deformation Δ with respect to the design shape as the difference.

[0050] In step **108** of FIG. 2, CPU **81** determines whether the deformation Δ exceeds the first threshold $\Delta s1$. Then, if it is determined YES in step **108** of FIG. 2, the process proceeds to step **109** of FIG. 2, and the coolant flow rate Q at the time of the next casting is increased to a $Q2$ larger than $Q1$ of the initialization. On the other hand, if CPU **81** determines NO in step **108** of FIG. 2, it skips step **109** of FIG. 2 and proceeds to step **110** of FIG. 2. Then, in step **110** of FIG. 2, CPU **81** operates a robot (not shown) to remove the die-cast product **90** from the movable mold **30**.

[0051] In step **111** of FIG. 2, CPU **81** determines whether or not casting has been performed continuously a predetermined number of times. If CPU **81** determines YES in step **111** of FIG. 2, the casting process is stopped. On the other hand, if it is determined NO in step **111** of FIG. 2, CPU **81** returns to step **101** of FIG. 2 to begin the next casting.

[0052] Next, with reference to FIG. 4, a change in the deformation amount Δ and the coolant flow rate Q of the die-cast product **90** with respect to the number of times of continuous casting will be described. In FIG. 4, a solid line A indicates a change in the deformation amount Δ , and a broken line B indicates a change in the coolant flow rate Q .

[0053] As shown in FIG. 4, when the first casting is performed, the coolant flow rate Q is a default $Q1$. When the first casting is completed, the temperature of the die-cast product **90** at the end of the casting is not very high, and the deformation amount Δ is also small.

[0054] As the number of times of continuous casting increases, the temperature of the die-cast product **90** at the end of casting gradually increases, and as shown by the solid line A in FIG. 4, the deformation amount Δ gradually increases. Then, during the N -th casting, the deformation amount Δ exceeds the first threshold value $\Delta s1$. At this time, as indicated by a broken line in FIG. 3, the outer peripheral portions **91** and **92** of the die-cast product **90** are deformed. Then, CPU **81** determines YES in step **108** of FIG. 2, and increases the coolant flow rate Q to a $Q2$ larger than the

default Q1 in step 109 of FIG. 2.

[0055] At the time of the N+1-th casting, CPU 81 makes the duty of the coolant pump 36 larger than the initial setting in step 103 of FIG. 2, and makes the coolant flow rate Q of the coolant channel 32 Q2 (see the broken line B in FIG. 4). As a result, in the N+1-th casting, the temperature of the die-cast product 90 is lower than that in the N-th casting when the mold clamping holding is performed in step 104 of FIG. 2. Therefore, as shown by the solid line A in FIG. 4, in the N+1-th casting, the deformation amount Δ is smaller than the first threshold value $\Delta s1$.

[0056] Thereafter, as the number of times of continuous casting increases, the deformation amount Δ increases again. CPU 81 again increases the coolant flow rate Q when the deformation Δ exceeds the first threshold $\Delta s1$.

[0057] As described above, in the casting apparatus 100, when the deformation Δ exceeds a predetermined threshold, CPU 81 increases the coolant flow rate Q. Therefore, an increase in the temperature of the die-cast product 90 during continuous casting can be suppressed, and an increase in the deformation amount Δ of the die-cast product 90 can be suppressed.

[0058] Next, with reference to FIG. 5 and FIG. 6, another operation of the casting

[0059] apparatus 100 when casting the die-cast product 90 will be described. Another operation is an operation of increasing the mold clamping holding time T instead of increasing the coolant flow rate Q when the deformation amount Δ exceeds the second threshold value $\Delta s2$. The same operations as those described above with reference to FIG. 2 are denoted by the same reference numerals, and the description thereof will be omitted. The second threshold value $\Delta s2$ is another threshold value different from the first threshold value $\Delta s1$.

[0060] As shown in FIG. 5, in another operation, CPU 81 determines whether the deformation Δ exceeds the second threshold $\Delta s2$ in step 201 of FIG. 5. If CPU 81 determines YES in step 201 of FIG. 5, the process proceeds to step 202 of FIG. 5, and the mold clamping holding time T at the time of the next casting is set to T2 longer than the default T1.

[0061] With reference to FIG. 6, the variation of the deformation amount Δ and the mold clamping holding time T of the die-cast product 90 with respect to the number of times of continuous casting in another operation will be described. In FIG. 6, a solid line C indicates a change in the deformation amount Δ , and a dashed-dotted line D indicates a change in the mold clamping holding time T.

[0062] As shown in FIG. 6, when the deformation amount Δ exceeds the second threshold $\Delta s2$ at the number of times of casting of the M-th time, CPU 81 sets the mold clamping holding time T to T2 longer than the default T1 in step 202 of FIG. 5 (see the dashed-dotted line D in FIG. 6). Thus, in the M+1-th casting, CPU 81 holds the mold clamping for T2 longer than the default T1 by the mold clamping device 40, so that the temperature of the die-cast product 90 is lower than that during the M-th casting. Therefore, as shown by the solid line C in FIG. 6, in the M+1-th casting, the deformation amount Δ is smaller than the second threshold value $\Delta s2$. As a result, it is possible to suppress an increase in the deformation amount Δ of the die-cast product 90 during continuous casting.

[0063] In the above explanation, in other operations, CPU 81 has been explained as increasing the mold clamping holding time T instead of increasing the coolant flow rate Q. However, without being limited thereto, CPU 81 may be performed by combining an increased coolant flow rate Q and an extended mold clamping holding time T.

[0064] Further, in the above description, the outer shape of the die-cast product 90 is detected by the laser scanners 71 and 72. However, the present disclosure is not limited thereto, and for example, an ultrasonic shape detector that detects the outer shape of the die-cast product 90 by ultrasonic waves may be used. As a result, the outer shape of the die-cast product 90 can be detected in a non-contact manner.

[0065] In the above description, the coolant channel 32 is provided in the insert 35 of the movable mold 30, but the present disclosure is not limited thereto. For example, a coolant channel may be

provided in the insert 25 of the fixed mold 20 to allow the coolant to flow through the coolant pump 36. In addition, a coolant channel may be provided in both the insert 35 of the movable mold 30 and the insert 25 of the fixed mold 20 to allow the coolant to flow therethrough.

[0066] Each step of the casting apparatus 100 described with reference to FIG. 2 is also a casting method performed by the casting apparatus 100. In this case, steps 105 and 106 of FIG. 2 are steps of detecting the outer shape of the die-cast product 90 by the laser scanners 71 and 72 when the mold 10 is separated after casting. Step 107 of FIG. 2 is a step of calculating the deformation amount Δ with respect to the design shape of the die-cast product 90 based on the outer shape of the die-cast product 90. Further, steps 108 to 109 of FIG. 2 are steps of increasing the coolant flow rate Q of the coolant pump 36 when the deformation amount Δ exceeds a predetermined threshold value. Steps 201 to 202 in FIG. 5 are steps of increasing the mold clamping holding time T when the deformation amount Δ exceeds a predetermined threshold value.

Claims

1. A casting apparatus that performs casting of a die-cast product, the casting apparatus comprising: a mold made up of a movable mold and a fixed mold, within which is provided a coolant channel; a mold clamping device for performing mold clamping and separation of the mold; a coolant pump for causing coolant to flow over the coolant channel of the mold; a shape detector for detecting an external shape of the die-cast product; and a control unit for adjusting mold clamping holding time of the mold by the mold clamping device or a coolant flow rate of the coolant pump, wherein the control unit includes a processor for performing information processing, and the processor acquires the external shape of the die-cast product from the shape detector when separating the mold following casting, calculates a deformation amount with respect to a designed shape of the die-cast product, based on the external shape of the die-cast product that is acquired, and adjusts the mold clamping holding time or the coolant flow rate based on the deformation amount that is calculated.
 2. The casting apparatus according to claim 1, wherein the processor increases the coolant flow rate when the deformation amount that is calculated exceeds a predetermined threshold value.
 3. The casting apparatus according to claim 2, wherein the processor increases the mold clamping holding time when the deformation amount that is calculated exceeds a predetermined other threshold value.
 4. A casting apparatus according to claim 1, wherein the shape detector is a laser scanner or an ultrasonic shape detector.
 5. A casting method of a die-cast product, the casting method comprising: preparing a casting apparatus that includes a mold made up of a movable mold and a fixed mold, within which is provided a coolant channel, a mold clamping device for performing mold clamping and separation of the mold, a coolant pump for causing coolant to flow over the coolant channel of the mold, and a shape detector for detecting an external shape of the die-cast product; detecting the external shape of the die-cast product by the shape detector when separating the mold following casting; calculating a deformation amount with respect to a designed shape of the die-cast product, based on the external shape of the die-cast product that is detected; and increasing a coolant flow rate of the coolant pump when the deformation amount that is calculated exceeds a predetermined threshold value, or increasing the mold clamping holding time of the mold when the deformation amount that is calculated exceeds a predetermined other threshold value.
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